5 **(a)** State two differences between a progressive wave and a stationary wave.

....

....

....

.... [2]

(b) Two coherent radiowave sources S_1 and S_2 that are in phase face each other, separated by a distance of 1.25 m, as shown in Fig. 5.1. Point P is located along line AB, *x* metres away from A.

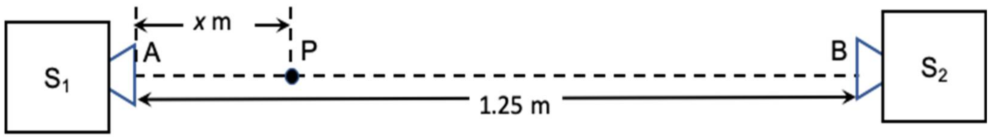

Fig. 5.1

(i) Show that the positions of minimum intensity along the line AB can be described by the following equation:

$$2L = (2n+1)\lambda$$

where
L is the path difference between the waves from S_1 and S_2,
n is any integer and
λ is the wavelength of the radiowaves.

[1]

(ii) Determine the number of points along AB where minima would be expected when radiowave signals of wavelength 0.429 m are produced.

number of points = ... [3]